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Dosimetry, half-life and statistical analysis of radioactive decays

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In the experiment the local equivalent dose rates from a Cs137-sample were measured in the laboratory and at various distances. We investigated responsivity of the Geiger counter to γ-radiation and determined it to be approximately 0.5%. The half-life of Ba-137m was found to be $T_{1/2} = (142.5 \pm 1.9)s$. Studying the stochastic nature of radioactive decay we performed χ^2-tests to analyze the distribution of decay events from a tile coated with uranium trioxide for gate times $t_{\text{gate}} \in \{1\,\text{s}; 10\,\text{s}\}$. The hypothesis that the data follow a Poisson distribution could not be rejected at a 5% significance level. At the same significance level the assumption that the Poisson distribution for $t_{\text{gate}} = 10\,\text{s}$ can be approximated by a normal distribution was rejected .

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1 Introduction

Radioactivity refers to the property of unstable atomic nuclei to spontaneously emit ionizing radiation into their surroundings. The emitted radiation — typically in the form of radioactive radiation — can excite atoms or molecules as it passes through matter, transferring energy in the process. Such energy transfer may displace atoms spatially, eject electrons from their atomic shells, or break chemical bonds. For this reason, it is essential to take appropriate protective measures and exercise caution when working with radioactive materials. Unlike other types of radiation, such as heat, the effects of ionizing radiation are not immediately perceptible, which increases the potential danger.

Despite these hazards, ionizing radiation also has many beneficial applications. In medicine, it can be used to specifically destroy cancer cells. Furthermore, it is utilized in nuclear reactors for energy production.

This paper examines the fundamental properties of radioactive decay, including the concept of half-life, and presents further investigations using a Geiger counter.

2 Theoretical backround

2.1 Law of radioactive decay

Radioactive decay is a stochastic process, meaning it is inherently random and it is not possible to predict the exact moment when a particular atom will decay. However, it is possible to calculate the probability of decay occurring within a given time interval, which enables an accurate statistical description of the behavior of a large number of identical atoms [2].

If we begin with an initial number N_0 of undecayed atoms, the number of atoms $N(t)$ remaining at a later time t follows the decay law:

$$N(t) = N_0 \cdot e^{-\lambda t}, \tag{1}$$

where λ is the decay constant [2]. The decay constant is related to the half-life $T_{1/2}$—the time it takes for half of the original nuclei to decay—by the equation:

$$\lambda = \frac{\ln(2)}{T_{1/2}}. \tag{2}$$

The activity $A(t)$, defined as the rate of decay (i.e., the number of decays per second), is given by the time derivative of $N(t)$:

$$A(t) = -\frac{dN(t)}{dt} = \lambda N(t). \tag{3}$$

Its unit is Becquerel (Bq), where one Becquerel represents one nuclear decay per second. It also follows an exponential decay behavior, analogous to $N(t)$.

2.2 Functionality of a Geiger-Müller Counter

A Geiger-Müller (GM) counter is a radiation detection device that operates based on the ionization of gas molecules by incident radiation. The core component of a GM counter is the Geiger-Müller tube, which consists of a cylindrical cathode and a thin axial anode wire enclosed in a gas-filled chamber, typically filled with an inert gas such as argon at low pressure, often with a halogen or organic vapor as a quenching agent.

When ionizing radiation (e.g. radioactive radiation) enters the tube, it interacts with the gas, ionizing some of its atoms and producing free electrons and positive ions. Due to the high voltage applied between the anode and cathode (typically between 400–900 V), the primary electrons are accelerated toward the anode wire, gaining sufficient energy to ionize additional gas atoms through collisions. This leads to a cascade of secondary ionizations, known as a Townsend avalanche, resulting in a large number of charge carriers from a single ionizing event.

This avalanche produces a measurable current pulse, which is then counted as a single detection event. The size of the output pulse is independent of the energy of the incident particle, which means the GM counter can detect the presence of radiation but does not provide spectroscopic information.

To prevent continuous discharge and allow the detector to return to its initial state, a quenching process is required. Quenching gases or organic vapors absorb the excess energy of positive ions or photons that might otherwise cause secondary avalanches.

The GM counter is highly sensitive and can detect single particles, but its dead time (typically $\sim 100\,\mu s$) limits its ability to measure high count rates accurately.

3 Dosimetry and radiation exposure

Because of the possible damages the human body can experience after being exposed to a certain amount of ionizing radiation, its quantification is necessary. The most fundamental and widely used measure for the amount of radiation absorbed is the *absorbed dose* D, defined as:

$$D = \frac{dE_D}{dm}, \tag{4}$$

where dE_D represents the amount of energy deposited by the radiation in a small mass element dm of the absorbing material [3]. The absorbed dose provides a direct measure of the energy absorbed per unit mass and is typically expressed in grays (Gy), where 1 Gy = 1 J/kg.

Since the interaction of different kinds of radiation with biological tissue differs, the *equivalent dose* H is introduced to account those differences. It is a weighted version of the absorbed dose that reflects the biological effectiveness of the radiation. It is typically expressed in sievert (Sv), where 1 Sv = 1 J/kg. The equivalent dose is given by:

$$H = q \cdot D \tag{5}$$

The weighting factor q depends on the type and energy of the radiation. For instance, $q = 1$ for gamma and beta radiation, while for low-energy alpha radiation, which causes more severe localized damage, $q = 20$. These values are chosen such that the equivalent dose H better reflects the probability of stochastic health effects, such as cancer, occurring as a result of radiation exposure.

For safety reasons, both the dose rate and the cumulative equivalent dose were monitored throughout the duration of the experiment. The threshold for acceptable exposure varies depending on individual health conditions, but according to safety standards, the annual dose should not exceed 5 mSv [3].

To determine the radiation exposure in the laboratory, a Radex RD1503 dosimeter was used to measure the equivalent dose rate $\dot{H}$ of the ambient background radiation. Since only gamma and beta radiation were expected, a radiation weighting factor of $q = 1$ was applied in the evaluation:

$$\dot{H} = q \cdot \dot{D} \quad \text{with} \quad q = 1. \tag{6}$$

The total equivalent dose H for a person staying in the laboratory for a time intervall t can be calculated using:

$$H = t \cdot \dot{H}, \tag{7}$$

assuming the background radiation remains constant throughout the time intervall.

Being closely to the sample for four hours would lead to an exposure of

$$\dot{H}_{0m} = (17.7 \pm 1.7)\tfrac{\mu Sv}{h}$$
$$H_{0m,4h} = (70.7 \pm 6.8)\mu Sv$$

For radiation at a distance of $(100.0 \pm 0.5)cm$ from the sample we could measure

$$\dot{H}_{1m} = (0.280 \pm 0.034)\tfrac{\mu Sv}{h}$$
$$H_{1m,4h} = (1.12 \pm 0.14)\mu Sv$$

For backround radiation we measured the dose rate: $\dot{H}_{background} = (0.1118 \pm 0.0058)\frac{\mu Sv}{h}$. The obtained equivalent dose of backround radiation for a person staying in the laboratory for a whole year is: $H_{background,1a} = (980 \pm 51)\mu Sv$

This value corresponds to approximately 20% of the permitted annual dose limit of $H_{\max,1\,\mathrm{a}} = 5$ mSv [3], indicating that the background radiation exposure in the laboratory does not pose a significant health risk.

4 Activity of the Cs137 sample

The activity A describes the decay rate of a radioactive sample and thus decreases with time t. It follows the exponential decay law:

$$A(t) = A_0 \cdot e^{-\lambda t}, \tag{8}$$

where A_0 is the initial activity and λ is the decay constant. The decay constant is related to the half-life $T_{1/2}$ by:

$$\lambda = \frac{\ln(2)}{T_{1/2}}. \tag{9}$$

For the isotope under consideration, the half-life is given as $T_{1/2} = 30.2a$. From the inventory map we know the sample is 46.75a old and had a starting activity of $A = 3.7MBq$. So the sample we used had a estimated activity of $A = 1.27MBq$ at that time.

5 Analysis of counting statistics of radioactive decays

The decay law in equation 1 gives the expected value for the remaining number of undecayed atoms. In a real observation the actually number will fluctuate around this value because of the stochastic nature of radioactive decay. A closer examination of the underlying statistics is necessary to accurately analyze data describing radioactive processes.

During a given time interval t_{Gate}, only an integer number of decays can be observed. The probability of a decay follows a binomial distribution with the number of atoms N_{atom} and the probability p_{decay} for one atom to decay during the observation period.

In the limit where $N_{\mathrm{atom}} \to \infty$ and $p_{\mathrm{decay}} \to 0$, while keeping the mean $\mu = N_{\mathrm{atom}} \cdot p_{\mathrm{decay}}$ constant, the binomial distribution approaches the Poisson distribution. The probability of observing k decays is then given by:

$$P(k) = \frac{\mu^k}{k!} e^{-\mu}. \tag{10}$$

For sufficiently large mean values μ, the Poisson distribution can be further approximated by a normal distribution with standard deviation $\sigma = \sqrt{\mu}$. The probability density function is:

$$P(k) = \frac{1}{\sqrt{2\pi\mu}} \exp\left(-\frac{(k-\mu)^2}{2\mu}\right). \tag{11}$$

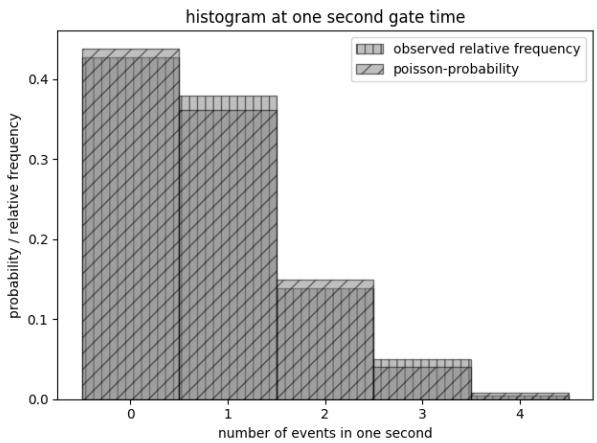

Figure 1: Distribution of the number of counts detected in a $t_{gate} = 1s$ interval. A Poisson distribution was fitted using the mean value of the experimental data.

In the following experiment we used the Leybold counter and a tile coated with uranium trioxide. With that a series of measurements is performed for gate times $t_{\mathrm{gate}} = 1\,\mathrm{s}$ and $t_{\mathrm{gate}} = 10\,\mathrm{s}$ where we examined the quality of these approximations. The theoretical expectation of a Poisson distribution, calculated with the mean value μ, is compared to the experimental data in Fig. 1 for $t_{\mathrm{gate}} = 1\,\mathrm{s}$.

Since measurements with a larger gate time result in a higher mean value, a normal distribution with standard deviation $\sigma = \sqrt{\mu}$ is additionally fitted to the data. The result of this fit is shown in Fig.2.

A χ^2-test is performed, so there is a quantification of the quality of the approximation as a Poisson distribution. The tested null hypothesis is that the data is Poisson distributed at a significance level of $\alpha = 5\%$.

5.1 Gate time $t_{gate} = 1s$

For the gate time $t_{gate} = 1s$ we got the mean value $\mu = 0.824 \pm 0.028$ and calculated $\chi^2 = 2.358$. The critical χ^2 at our significance level of 5% is $\chi^2_{crit} = 7.815$. From that we can state, that the data follows a poisson distribution on our given significance level.

5.2 Gate time $t_{gate} = 10s$

For the gate time $t_{gate} = 10s$ we obtained the mean value $\mu = 8.287 \pm 0.091$ and calculated $\chi^2 = 12.774$ for the poisson distribution while the critical χ^2 is $\chi^2_{crit} = 22.362$. For the gaussian distribution we got $\chi^2 = 30.948$ and $\chi^2_{crit} = 21.026$. From that we can state, that the data follows a poisson but not a gaussian distribution on our given significance level.

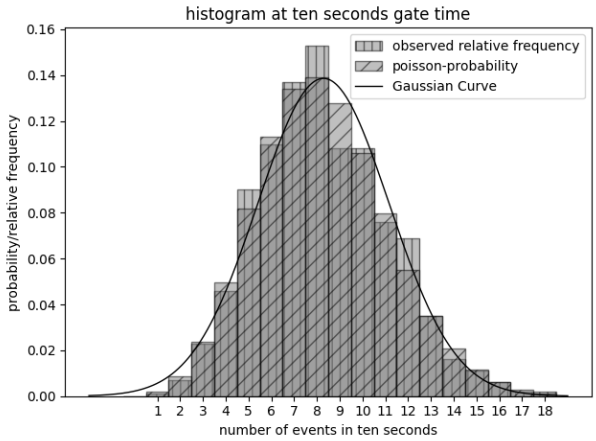

Figure 2: Distribution of the number of counts detected in a $t_{gate} = 10s$ interval. A Poisson and a normal distribution were fitted using the mean value of the experimental data. For the Gauss fit the standard deviation was set to be the expected value $\sigma = \sqrt{\mu}$, assuming the data follows a Poisson distribution.

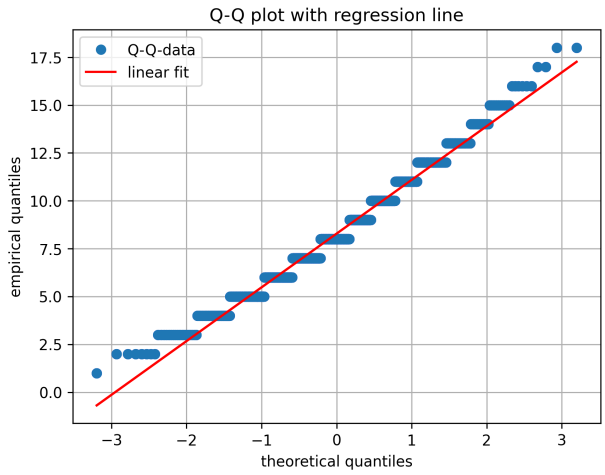

Figure 3: Q-Q plot of a gaussion distribution for the 10s data shows a systematic error

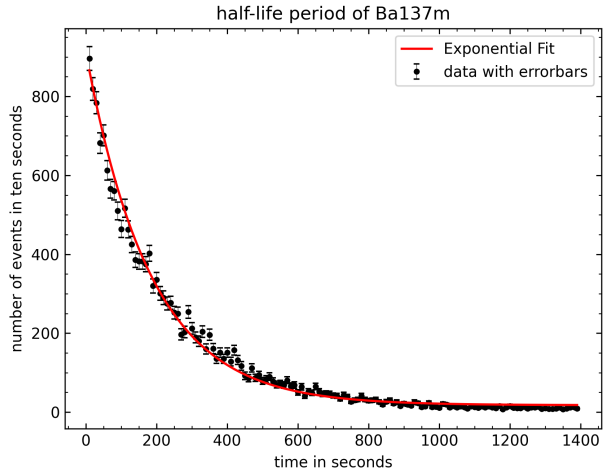

Figure 4: Decay rate of a Ba137m sample with an exponential fit. The exponent of the fit can be used to calculate the half life of Ba137m as $T_{1/2} = (142.5 \pm 1.9)s$.

In addition to that we made a Q-Q-Plot for a gaussian distribution, which shows a systematic error, so we should not assume a gaussian distribution.

6 Counting tube characteristics

6.1 Measuring the null effect with the counter tube

Due to cosmic rays and naturally occurring radioactive isotopes there is always a non-zero radiation level left even in the absence of specific radioactive materials. That means an unshielded Geiger counter will always register some events. We have to account this background radiation when determining the activity of a particular radioactive sample.

To quantify this so called null effect, measurements were performed with the Leybold counter both on the laboratory workbench and inside a lead shielding cube. The number of detected decays was recorded over a time interval of $t_{\text{gate}} = 5\,\text{min}$. With lead shielding there were 20 events per minute and without lead shielding there were 38 events per minute detected. The shielding leads to a reduction of the null effect by $(0.479 \pm 0.038)\%$

6.2 Half-life period of Ba137m

For measuring the half-life of Ba137m we used a sample of Cs137 which can decay by emitting β-radiation both into the ground state and the excited state of Ba137. The excited state Ba137m then decays into ground state emitting γ-quanta as shown in Fig. 5. Due to the different solubility of the two elements, Ba137 can be separated from Cs137 and then put in front of the Leybold Geiger counter to measure the

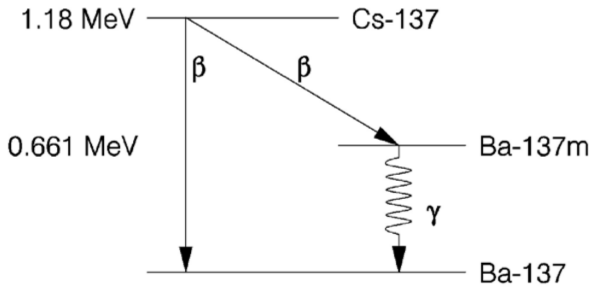

Figure 5: Decay scheme for Cs137. The isotope decays either into an excited or into the ground state of Ba137.

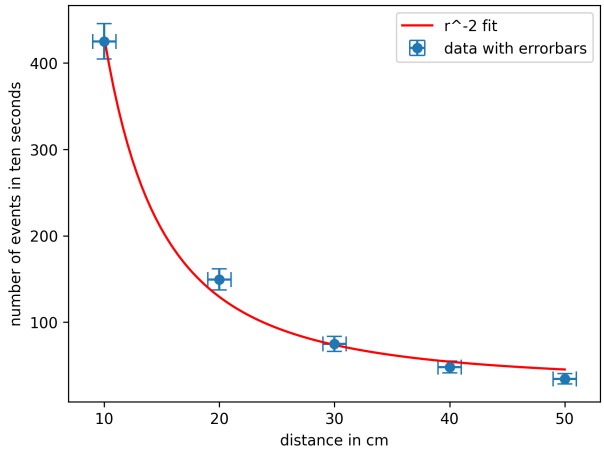

Figure 6: Plot of count rates against the inverse square of the distance. For each distance, five measurements were performed using a gate time of $t_{\text{gate}} = 10s$

number of detected decays within the gate time of $t_{gate} = 10s$. Fig. 4 shows the resulting data and an exponential fit according to

$$A(t) = a \cdot e^{-t/b} + c \tag{12}$$

where c accounts for the background (null effect). The fit parameter b allows us to calculate the half-life according to Eq. (2) using $\lambda = \frac{1}{b}$. We get a half-life of $T_{1/2} = (142.5 \pm 1.9)s$ which is about 25s shorter than the literature value of $T_{1/2,\text{lit}} = 153.27s$ [4]

6.3 Response capacity of the counter tube for gamma-quanta

While measuring radioactive radiation with a counter tube, not every incoming γ-quantum interacts with the filling gas inside the tube because a counter tube has a finite extent. The ratio between the number of incoming and the number of detected γ-quanta is called the *responsivity* R_γ of the counter to γ-radiation. In this part of the experiment, measurements were taken at five different distances from the

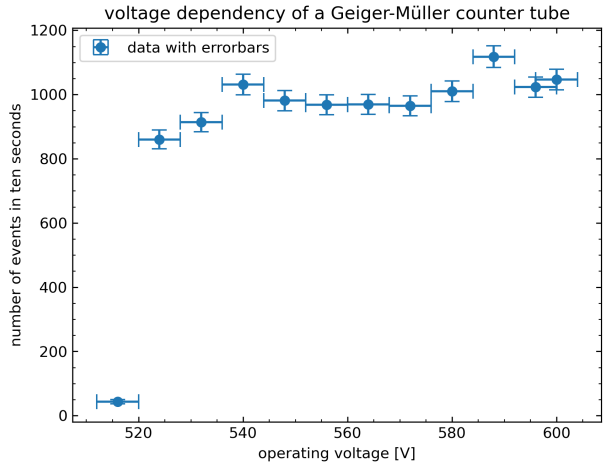

Figure 7: Characteristic curve of a Geiger-counter. For voltages below the threshold voltage of $U_{threshold} = (516. \pm 4.)$ V it does not detect any radiation. Above this voltage the countrate increases rapidly and reaches a plateau, meaning the countrate does not depend on the exact voltage.

radioactive source: 10, 20, 30, 40, and 50 cm. For each distance, five measurements were performed using a gate time of $t_{\text{gate}} = 10s$. We assume that the emitted radiation spreads uniformly in all directions. To verify this assumption, the count rates were plotted against the inverse square of the distance as shown in Fig. 6, which yielded a linear fit, confirming the expected geometric behavior of the radiation.

For each distance, the responsivity R of the detector was calculated using the following equation:

$$R\,(\%) = 100 \cdot \frac{A_{\text{sphere}}(r) \cdot C_{\text{detected}}}{A_{\text{detector}} \cdot C_{\text{theoretical}}}, \tag{13}$$

where $A_{\text{sphere}}(r) = 4\pi r^2$ is the surface area of a sphere at distance r, C_{detected} is the measured count rate, A_{detector} is the effective detection area of the Geiger-Müller counter, and $C_{\text{theoretical}}$ is the theoretical activity of the source (1.27 MBq).

The average responsivity and its standard error were determined from the individual values of R obtained at each distance. The result is

$$R_\gamma = (0.534 \pm 0.044)\%$$

6.4 Characteristics of the used Geiger-Müller counter

We place the opening of the Cs137 sample in front of the counter tube and measure the detected counts during a gate time of $t_{\text{gate}} = 60$ s for different tube voltages. The results are shown in Fig.7 and follow the discussed theory. The threshold voltage is determined as $U_{\text{threshold}} = (516(4)\,\text{V})$.

For higher voltages, the count rate appears to increase slowly, which contradicts the theoretical expectation. We suspect that this apparent slope is actually caused by statistical scattering around the mean value, which should ideally remain constant in this plateau region.

To further investigate this behavior, it would be necessary to perform a series of repeated measurements at each voltage setting to assess the statistical variation more accurately.

7 Conclusion

We first measured the equivalent dose rate for background radiation and at various distances from a Cs-137 sample. From that we can state that the exposure to background radiation in the laboratory does not pose a significant health risk, as a person spending one year in the lab would not exceed the annual dose limit of $H_{\mathrm{max,1a}} = 5\mathrm{mSv}$.

The background count rate of the Geiger counter was measured to be $38min^{-1}$ and with lead shielding that could be reduced to $20min^{-1}$. The responsivity of the Geiger counter to γ-quanta was determined to be $R_\gamma = (0.534 \pm 0.044)\%$ using a Cs-137 sample of known activity.

The half-life of Ba-137m was found to be $T_{1/2} = (142.5 \pm 1.9)s$, in which does not agree with the literature value within the margin of error.

We statistically analyzed the distribution of radioactive decays by performing χ^2 tests to assess whether the data follows a Poisson distribution. This assumption could not be rejected at a 5% significance level for gate times $t_{\mathrm{gate}} \in \{1\,\mathrm{s}, 10\,\mathrm{s}\}$. The assumption that this Poisson distribution can be approximated by a normal distribution for $t_{\mathrm{gate}} = 10\,\mathrm{s}$ was rejected at the same significance level.

References

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8 Appendix